# Relational vs. Graph Databases

## Technical Comparison

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# Outline

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- Storage Layout
- Query Complexity Analysis
- Experimental Analysis: Recipes Dataset
- Query Language Analysis
- Conclusions

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# Storage Layout



## Row / Columnar Store

**Row-oriented (OLTP)** — tuple contiguous in page:

| id | name | age | city |
|---|---|---|---|
| 1 | Alice | 30 | Rome |
| 2 | Bob | 25 | Milan |

**Column-oriented (OLAP)** — one array per column:

```
age[]:    [30, 25, 41, ...]
city[]:   [Rome, Milan, ...]
```

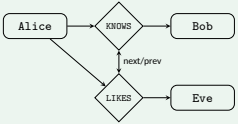
- Compression ratio ↑ on repeated values
- Analytics scan only needed columns
- Examples: Redshift, DuckDB, Parquet

## Native Graph Storage (Neo4j)

**Fixed-size records on disk:**

- Node record: **15 B** Rel record: **34 B**

**Index-free adjacency** — each node holds a pointer to its *first* relationship; rels form a doubly-linked list:




- Each node holds a **direct pointer** to its first relationship record
- Properties in a **separate property store**

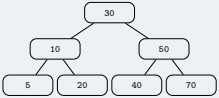
# Index Structures

1 Storage Layout

## Relational DB — B-Tree Index

**B-Tree** (default in PostgreSQL, MySQL):

- Balanced tree over a **column's values**
- Lookup, insert, delete: $O(\log n)$
- Supports equality *and* range queries




## Graph DB — Index-Free Adjacency

**No index needed for traversal:**

- Neighbour lookup: $O(1)$ — just follow a pointer
- Cost depends on **neighbourhood size**, not graph size

**Indexes still used for entry points:**

- **B-Tree / Hash** — find node by property value
- **Full-text (Lucene)** — text search on properties

RDB: index resolves **every JOIN** — $O(\log n)$ per join, globally.
Graph DB: index only for the **entry point** — hops are $O(1)$ pointers.

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# Query Complexity: Multi-Hop Joins



## Relational

$k$-hop query requires $k$ self-JOINs on the edge table.

Assuming hash-join, each JOIN costs $O(n)$ where $n = |E|$.

**Total**: $O(k \cdot n)$ — grows with graph size, not neighbourhood size.

With index on foreign keys: $O(k \cdot d)$ per lookup ($d$ = avg degree), but planning overhead for large $k$ remains.

## Graph

Traversal cost proportional to **neighbourhood size**, not $|V|$ or $|E|$.

$k$-hop traversal: $O(d^k)$ where $d$ is average degree.

**For sparse graphs** ($d \ll n$): dramatically faster.

**For dense super-nodes** (e.g. hubs): same bottleneck. Strategies: relationship type filtering, degree caps.

# Query Complexity: Aggregations & Analytics



## Relational — Aggregation Strength

- `GROUP BY` highly optimised (sort-merge, hash-agg)
- Window functions (`OVER PARTITION BY`) efficient
- Columnar layout ⇒ vectorised execution
- Cost-based optimiser rewrites complex queries
- Materialised views for pre-aggregation

**Best for**: OLAP, reporting, large flat aggregations

## Graph — Aggregation Limitations

- `COUNT`, `SUM`, `COLLECT` available in Cypher
- Less mature query optimiser than mature RDBMS
- No columnar layout ⇒ row-at-a-time aggregation
- Graph algorithms (PageRank, BFS, community detection) via **GDS library**
- APOC procedures extend analytical power

**Best for**: topology-aware aggregations (centrality, clustering)

# Complexity Summary Table



| Query Type | Relational | Graph DB |
|---|---|---|
| **Point lookup (PK)** | $O(\log n)$ | $O(1)$ |
| **1-hop neighbour** | $O(\log n)$ + JOIN | $O(d)$ |
| $k$**-hop traversal** | $O(k \cdot n)$ | $O(d^k)$ |
| **Shortest path** | $O(n^2)$ | $O((\lvert V\rvert + \lvert E\rvert) \log \lvert V\rvert)$ |
| **Global aggregation** | $O(n)$ | $O(n)$ |

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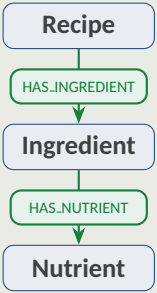
# Dataset Description

3 Experimental Analysis: Recipes Dataset

## Graph Schema




## Dataset Sizes Tested

| Scale | Recipes | Ingredients | Nutrients | R-I Edges | I-N Edges |
|---|---|---|---|---|---|
| 25% | ∼42.5k | 4805 | 142 | ∼373k | ∼304k |
| 50% | ∼85k | 4869 | 142 | ∼746k | ∼308k |
| 75% | ∼127.5k | 4870 | 142 | ∼1.12M | ∼308k |
| 100% | ∼170k | 4870 | 142 | ∼1.5M | ∼308k |

# Queries Tested



## 10 queries across 4 categories

**Multi-hop** (2-hop traversals):

- Nutrients for a recipe
- Recipes sharing an ingredient
- Filter mid-traversal by nutrient amount

**Aggregation**:

- Total nutrient in recipe
- Top ingredients by nutrient
- Recipes by prep time

**Mixed**:

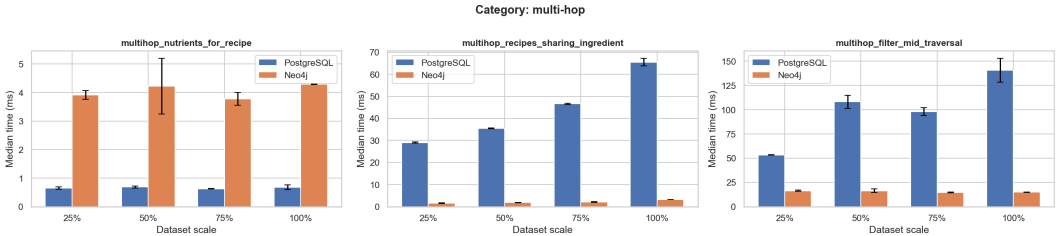
- Most used ingredients
- Subset matching given recipe
- Nutritional density

**Variable-length**:

- Reachable recipes in $N$-hops

# Results: multihop queries results

## 3 Experimental Analysis: Recipes Dataset



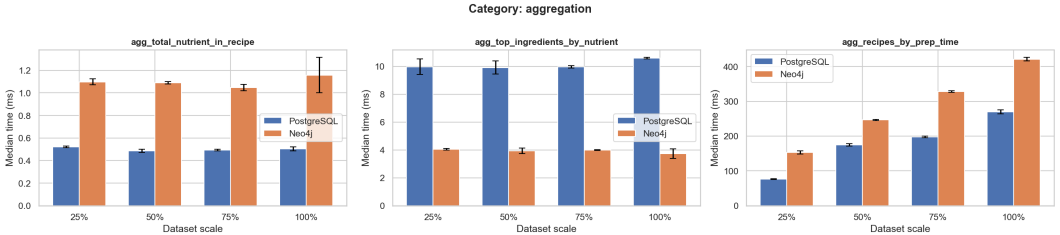
# Results: multihop queries analysis



- **Nutrients for a recipe**: shallow traversal, pattern `Recipe → Ingredient → Nutrient` with specific starting recipe. Very fast queries in both cases, we are measuring the fixed per-query overhead of the two DBs, Postgres is faster.
- **Recipes sharing an ingredient**: we notice the dramatic win of Neo4j as expected, the pattern `Recipe → Ingredient ← Recipe` generates very large intermediate results (fan-out) that grows with the dataset size, meanwhile in Neo4j simply follows the pointers.
- **Filter mid-traversal by nutrient amount**: the query is to find recipes containing an ingredient with more than X amount of a given nutrient. Pattern `Recipe → Ingredient → Nutrient` with specific nutrient and amount requires SQL to make a join of the two relations across the full dataset, in Neo4j it's possible to apply the filter on the amount during the traversal.

# Results: aggregation queries results

## 3 Experimental Analysis: Recipes Dataset



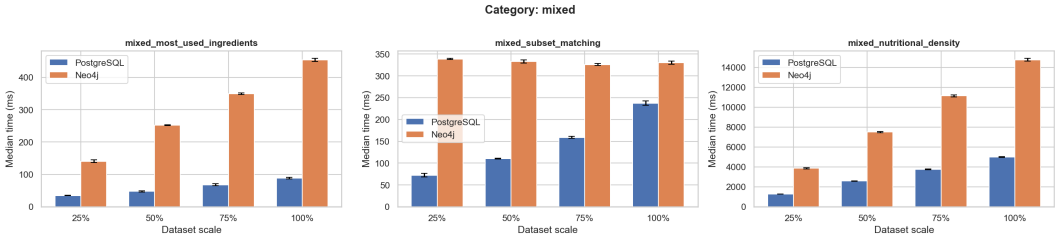
# Results: aggregation queries analysis



- **Total nutrients in recipe**: simple sum with selective filter on one recipe, again we are measuring the overhead.
- **Top ingredients by nutrient**: global aggregation that requires SQL to make a full scan of the `has_nutrient` relation filtered by `nutrient_id`, even with an index on `ingredient_id` produces a very large intermediate result, meanwhile in Neo4j a traversal starts from the `Nutrient` node and reaches all the connected Ingredient nodes, a sequential scan of the local neighborhood which is cache-friendly.
- **Recipes by prep time**: range filter on a property followed by a count. Postgres uses a B-tree index on `recipe.minutes` which is ideal for range queries. Neo4j instead is not optimized for property-based range filtering which require a full node scan of all the `Recipe` nodes.

# Results: mixed queries results

## 3 Experimental Analysis: Recipes Dataset



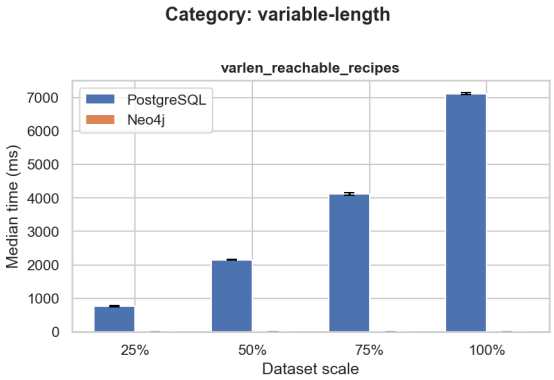
# Results: mixed queries analysis



- **Most used ingredients**: classic hash aggregation on the `has_ingredient` junction table, Postgres is extremely optimized for this, Neo4j instead must traverse every `Recipe → Ingredient` relationship in the graph without pruning.
- **Subset matching given recipe**: find recipes that contain all ingredients of a given recipe. Double `NOT EXISTS` query for postgres which is optimized for that using anti-joins. Neo4j must check every required ingredient ($N$ relationship existence) for each candidate recipe. We notice though that the computation time is constant for all dataset scales, which suggests that the cost may be dominated by a fixed overhead function (such as `collect()` or `ALL()`).
- **Nutritional density**: a 3-hop aggregation `Recipe→Ingredient→Nutrient` over the entire dataset with `AVG` and `COUNT DISTINCT`. Both systems are slow because the aggregation is global and the dataset is big, but Postgres wins because it uses hash aggregation with group-by on recipe ($O(N)$), and Neo4j must traverse every relation in the graph and the traversal step overhead of Neo4j is greater than the Postgres tuple comparison.

# Results: variable length query result





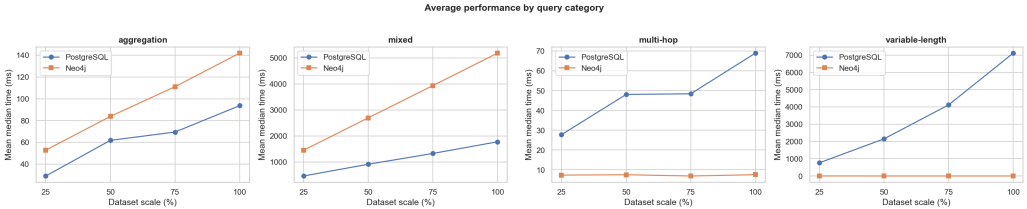
# Results: variable length query analysis



- **Reachable recipes in $N$-hops**: here Neo4j wins by orders of magnitude (sub ms vs 7000+ ms at full scale). This is the canonical showcase for graph dbs. The query counts distinct recipes reachable within $N$ (in this case 2 because otherwise Postgres was too slow) via shared ingredients.
  Postgres must implement BFS manually using a recursive CTE and each recursive step creates intermediate results which grow with data size.
  Neo4j has a **native variable-length path operator** (`HAS_INGREDIENT*1..N`) that maintains a compact visited set and follows relationship pointers without materializing intermediate results, pruining already visited nodes in $O(1)$.
  The reachability answer is bounded by graph topology and not global dataset size.

# Results: Average Performance by Query Category

## 3 Experimental Analysis: Recipes Dataset




**Neo4j worse** | **Neo4j much worse** | **Neo4j much better** | **Neo4j much better**

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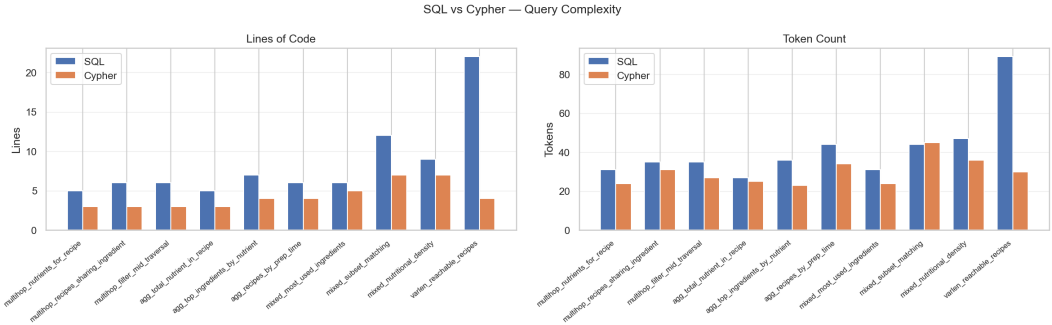
# Query verbosity: SQL vs. Cypher

## 4 Query Language Analysis



# Comparison



## Key differences

- **Relationships**: in SQL, relationships are expressed via foreign keys and JOINs, in Cypher, they have with their own type, direction, and properties and are easier to access (`-[:REL]->`)
- **Traversal**: multi-hop queries grow linearly in SQL (one JOIN per hop), Cypher handles them natively with variable-length patterns (`[:REL*1..n]`)
- **Verbosity**: SQL tends to be more verbose as query complexity grows, Cypher stays concise because the graph structure already encodes relationships

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# When to Choose a Relational Database



**Choose a Relational DB when:**

- Data is **tabular and well-structured** with a stable schema
- Workload is **transactional** (banking, e-commerce, ERP)
- Queries involve **global aggregations** or OLAP analytics
- Joins are **shallow** (1–2 levels)

**Choose a Graph DB when:**

- Data is **highly connected** and relationships are central
- Queries involve **multi-hop traversals** (fraud rings, social networks, supply chains)
- **Pathfinding** or **reachability** is required
- Schema evolves frequently (new relationship types)
- **Recommendation engines**
- **Knowledge graphs** or semantic data